

Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

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[varnikakomali@VARNIKAs-MacBook-Air Skill % nano skill_7.c
[varnikakomali@VARNIKAs-MacBook-Air Skill % gcc skill_7.c -o skill_7 && ./skill_7
=== Part 1: Command Resolution & PATH Verification ===

[Found]   ls           -> /bin/ls (Executable permission OK)
[Found]   echo         -> /bin/echo (Executable permission OK)
[Found]   whoami       -> /usr/bin/whoami (Executable permission OK)
[Missing] fakecmd_xyz  -> Command not found in PATH

=== Part 2: Process Synchronization & Monitoring with waitpid() ===

[Child 1 (PID: 66258)] Running task, exiting with code 42...
[Child 2 (PID: 66259)] Running task, raising SIGABRT...
[Parent] Monitored Child 1:
  - Exited Normally: Yes
  - Exit Status Code: 42
[Parent] Monitored Child 2:
  - Terminated By Signal: Yes
  - Signal Number: 6
varnikakomali@VARNIKAs-MacBook-Air Skill %
```